

Matrix:

1. Printing the matrix

```
import java.util.*;
```

```
class matrixprint{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int row= sc.nextInt();
```

```
        int col=sc.nextInt();
```

```
        int[][]matrix=new int[row][col];
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                matrix[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                System.out.print(matrix[i][j]+" ");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

OUTPUT: 2

2

1

2

3

4

1 2

3 4

2.Addition of 2 matrix:

```
import java.util.*;
```

```
class add{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int row = sc.nextInt();
```

```
        int col = sc.nextInt();
```

```
        int[][]matrix1=new int [row][col];
```

```
        int[][]matrix2=new int [row][col];
```

```
        int[][]sum=new int[row][col];
```

```
        System.out.println("Enter the 1st matrix");
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                matrix1[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        System.out.println("Enter the 2nd matrix");
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                matrix2[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                sum[i][j]=matrix1[i][j]+matrix2[i][j];
```

```
            }
```

```
        }
```

```
        System.out.println("The sum of two matrix:");
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                System.out.print(sum[i][j]+" ");
```

```

        }
        System.out.println();
    }
}

```

OUTPUT:2

2

Enter the 1st matrix:1

2

3

4

Enter the 2nd matrix:1

2

3

4

The sum of two matrix: 2 4

6 8

3.Matrix Transpose:

```
import java.util.*;
```

```
class trans{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int row= sc.nextInt();
```

```
        int col=sc.nextInt();
```

```
        int[][]matrix=new int[row][col];
```

```
        for(int i=0;i<row;i++){
```

```
            for(int j=0;j<col;j++){
```

```
                matrix[i][j]=sc.nextInt();
```

```
            }
```

```
        }
```

```
        System.out.println("The normal matrix is:");
```

```

for(int i=0;i<row;i++){
    for(int j=0;j<col;j++){
        System.out.print(matrix[i][j]+" ");
    }
    System.out.println();
}

```

```

System.out.println("The transpose matrix is:");
for(int i=0;i<row;i++){
    for(int j=0;j<col;j++){
        System.out.print(matrix[j][i]+" ");
    }
    System.out.println();
}
}

```

```

}

```

OUTPUT:

2

2

1

2

3

4

The normal matrix is:1 2

3 4

The transpose matrix is:1 3

2 4

4. Saddle point:

```
import java.util.Scanner;
```

```
public class SaddlePoint {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter rows and columns: ");  
        int r = sc.nextInt();  
        int c = sc.nextInt();  
  
        int[][] a = new int[r][c];  
  
        System.out.println("Enter matrix elements:");  
        for (int i = 0; i < r; i++) {  
            for (int j = 0; j < c; j++) {  
                a[i][j] = sc.nextInt();  
            }  
        }  
  
        boolean found = false;  
  
        for (int i = 0; i < r; i++) {  
            int minRow = a[i][0];  
            int colIndex = 0;  
  
            for (int j = 1; j < c; j++) {  
                if (a[i][j] < minRow) {  
                    minRow = a[i][j];  
                    colIndex = j;  
                }  
            }  
        }  
    }  
}
```

```

    }

    boolean isSaddle = true;
    for (int k = 0; k < r; k++) {
        if (a[k][colIndex] > minRow) {
            isSaddle = false;
            break;
        }
    }

    if (isSaddle) {
        System.out.println("Saddle Point: " + minRow +
            " at position (" + i + ", " + colIndex + ")");
        found = true;
    }
}

if (!found) {
    System.out.println("No saddle point found.");
}

sc.close();
}
}

```

OUTPUT:

Enter rows and columns: 3 3

Enter matrix elements:

1 2 3

4 5 6

7 8 9

Saddle Point: 7 at position (2,0)

5. Spiral order

```
import java.util.Scanner;
```

```
public class SpiralTraversal {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter rows and columns: ");  
        int r = sc.nextInt();  
        int c = sc.nextInt();  
  
        int[][] a = new int[r][c];  
  
        System.out.println("Enter matrix elements:");  
        for (int i = 0; i < r; i++) {  
            for (int j = 0; j < c; j++) {  
                a[i][j] = sc.nextInt();  
            }  
        }  
  
        int top = 0, bottom = r - 1;  
        int left = 0, right = c - 1;  
  
        System.out.println("Spiral Order:");  
  
        while (top <= bottom && left <= right) {  
            for (int i = left; i <= right; i++) {  
                System.out.print(a[top][i] + " ");  
            }  
            top++;  
            for (int i = top; i <= bottom; i++) {
```

```

        System.out.print(a[i][right] + " ");
    }
    right--;

    if (top <= bottom) {
        for (int i = right; i >= left; i--) {
            System.out.print(a[bottom][i] + " ");
        }
        bottom--;
    }

    if (left <= right) {
        for (int i = bottom; i >= top; i--) {
            System.out.print(a[i][left] + " ");
        }
        left++;
    }
}

sc.close();
}
}

```

OUTPUT:

Enter rows and columns: 3 3

Enter matrix elements:

1 2 3

4 5 6

7 8 9

Spiral Order:

1 2 3 6 9 8 7 4 5